



MINING DEALER BEST PRACTICE

Front Wheels Oil Drain Optimization for 777G OHT

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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May 2023
1.04-230510-1

1.0 Introduction

Maintenance efficiency is frequently optimized due to its importance. Thus, maintenance schedules, special tools, and several artifices are used to improve the physical availability of equipment and meet customer expectations, regarding product quality, workmanship, and environmental policies.

Considering this scenario, it was observed that, during the activity of draining the hydraulic oil from the front wheels, part of the oil was spilled on the wheel or even on the ground, contaminating the ground and leaving marks on the wheel during the operation of the equipment, once there was no suitable tool for such activity.

To carry out the task of draining the front wheels of the 777G fleet, field technicians used to place a bucket below the drain plug as Picture 1 illustrates. It does not stabilize on the wheel and eventually falls or moves out of position, causing oil to spill over the wheel and onto the ground. This entails, in addition to soil contamination, time loss during the preventive procedure, as the technician team needs to clean up the area affected by the oil in the equipment and on the floor.

In addition, the technician often needs to hold the bucket during the entire oil drain period to avoid spillage and contamination.



Picture 1 - Conventional oil collection.

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Picture 2 - The technician holding the bucket throughout the oil drain.

When the technician must hold the bucket to collect the oil, he loses about 40 minutes which he could be doing other tasks.



Picture 3 - Oil spillage on the wheel and on the workshop floor caused by the bucket falling during collection.

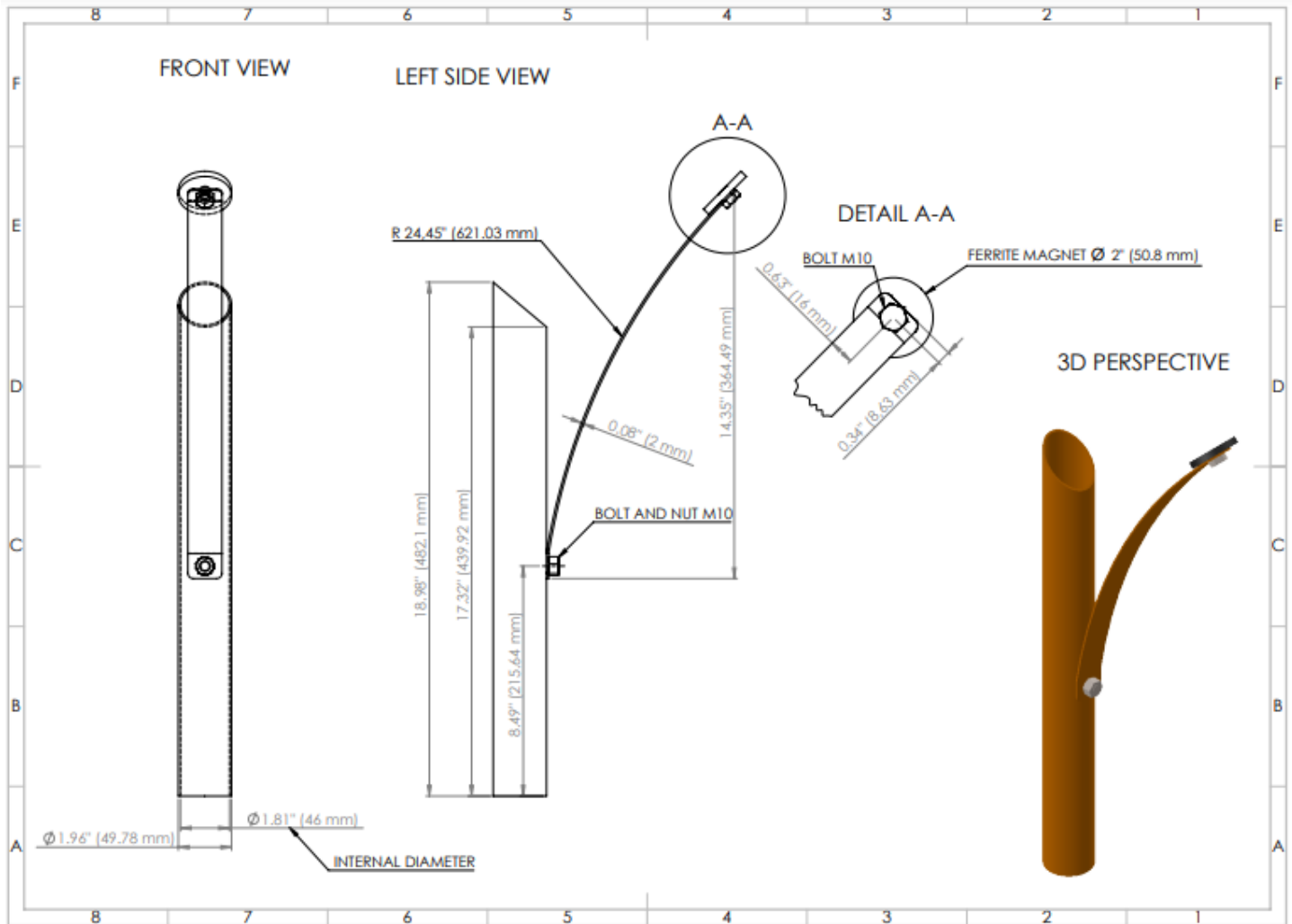
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2.0 Best Practice Description

Considering this circumstance, a tool capable to optimize the activity was developed, enabling effective oil draining, and avoiding time wasting and contamination on the equipment and the floor.

Pictures 2 to 5 show the tool schematic and its implementation on equipment.

This tool is focused on Off-Highway Trucks (OHT) 770-777.



Picture 4 - Tool drawing.

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Picture 5 - Drain tool.



Pictures 6 and 7 - Tool implementation.

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4. Construct drainage tool;



Picture 10 and 11 - Cut one end of the tube at the correct angle.



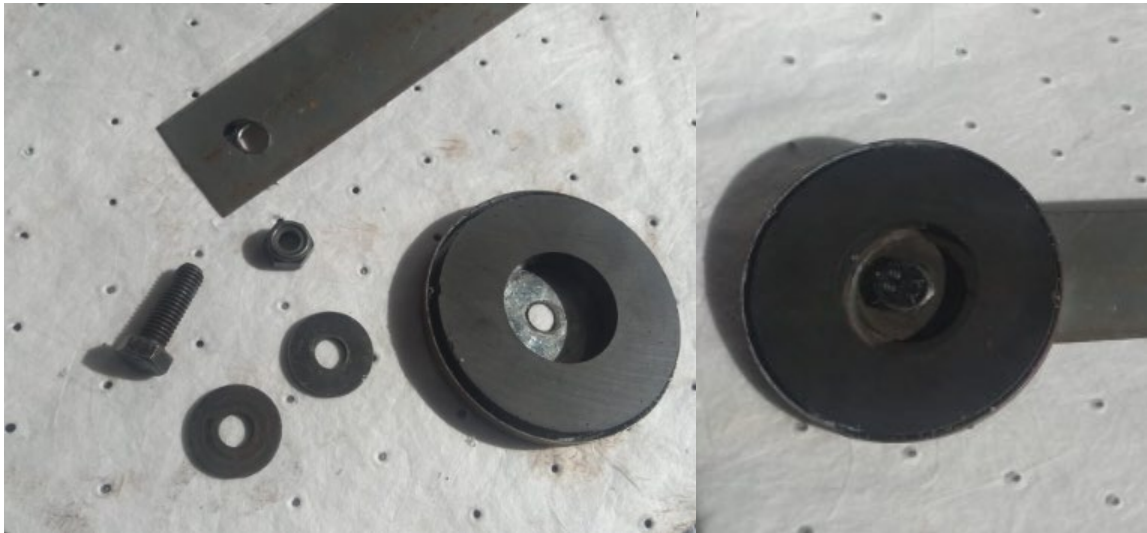
Picture 12 - Drill a hole in the tube to the size of 8,49" (215,64 mm).

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Picture 13, 14, and 15 - Drill a hole in both ends of the rod to 0,34" (8,63 mm).

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Picture 16 - Using the screw, nut, and washers, attach the magnet to one end of the rod.



Picture 17 - Position the screw with the head into the tube hole.

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Picture 18 - Bend the rod at the correct angle and attach it to the tube screw using the remaining nut and washer.



Picture 19 - Paint tool.

5. Test the tool in preventive activities.

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4.0 Benefits

- Reduction of preventive time;

In the event of an oil spill during collection, it was observed that the technician spent about two hours providing resources and cleaning the floor and wheels. With the tool, we started to avoid this type of event. in this way, we reduce preventive maintenance time by approximately two hours per piece of equipment.

In addition, the technician can carry out tasks while the oil is drained, without needing to hold buckets.

He gains about 40 minutes doing other preventive activities.

- Reduction of soil contamination.

5.0 Resources Required

- 1 PVC tube 49,78 ø x 482,1 - MM (1,96 ø X 18,98 inches) = US\$ 2,30.
- 1 Magnet 50,8 ø x 5 - MM (2 X 0,19 inches) = US\$ 3,56.
- 2 bolts (M10X1.5X20-MM) Part number 8T-4137 = US\$ 1,09.
- 3 washer (11X21X2.5-MM) Part number 8T-4121= US\$ 0,63.
- 2 nuts (M10X1.5-THD) Part number 344-5675 = US\$ 1,09.
- 1 Paint aerosol yellow Caterpillar = US\$ 11,46.
- 1 Metal rod 364,49 X 32 X 2- MM (14,35 X 1,26 X 0,08 inches) = US\$ 2,55.

The cost of the material may vary depending on the country where the tool is developed.

6.0 Supporting Attachments / References

See drawings.

7.0 Related Best Practices

NA.

8.0 Acknowledgements

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